

Please refer to **Appendix A** pp **A6-A9** of the recommended text book.

UNIT 1.2

THE ABSOLUTE VALUE

Learning objectives: On the completion of this unit you should

- (1) be able to write down and use the definition of the absolute value.
- (2) be able to write down and use the properties of the absolute value.
- (3) be able to solve inequalities with absolute values.
- (4) be able to sketch the graph of $y = |f(x)|$ from the graph of $y = f(x)$

Definition of absolute value. please see the definition on page A6

Examples: Express the following expressions without the absolute value symbol.

(1) $||-2| - |-3||$

$$(2) |3x - 2|$$

$$(3) |x - 2| \text{ if } x < 2$$

Geometric interpretation (meaning) of the absolute value.

Absolute values as functions.

- Discuss $f(x) = |x|$ as a function.

- Use the definition of the absolute value to write the following functions as piece-wise defined functions and sketch their graphs

(1) $f(x) = |x - 1|$

(2) $f(x) = |x^2 - 1|$

Properties of absolute values. Please refer to page A7 (These are crucial for solving equations and inequalities involving absolute values).

Examples: using properties of absolute value to solve equations and inequalities involving absolute values.

(1) Solve $|1 - x| = 0$.

(2) Solve $|2x - 3| = 2$.

(3) Solve $|3x - 4| \geq 2$.

(4) Solve $|5x - 2| \leq 1$.

(5) Solve $|x^2 - 9| = 1$.

(6) Solve $|1 - x| < 0$.

(7) Solve $|-x| < 0$.

(8) Solve $\left| \frac{x}{x-2} \right| > 0$.

(9) Solve $\frac{x}{|x-2|} \geq 0$.

(10) Solve $\frac{|x+1|}{|x-2|} > 0$.